

Hypothesis Testing I: Introduction

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Wednesday, Week 9

Goals for today

- Introduce hypothesis testing framework

Introduction

Motivating Example: Can some diseases be detected by smell?

Joy Milne, a retired nurse, claimed to have the ability to smell whether someone had Parkinson's disease. She noticed a change in her husband's smell 10 years before he was diagnosed with the disease, then recognized it again in support groups (this is a true story!).



Motivating Example: Can some diseases be detected by smell?

- Imagine we are researchers who want to test Joy's claim.
- We collect a bunch of tee shirts, half worn by individuals who are diagnosed with Parkinson's, half worn by individuals who aren't.
- We present Joy with tee shirts one at a time, and ask her to identify which group the wearer is in.
- **Discuss with Neighbor(s):**
  - **Q:** If Joy is guessing at random, what's the probability that she guesses correctly for **one** shirt?
  - **Q:** How many out of 10 randomly selected shirts would you expect her to guess correctly?
  - **Q:** What is the probability of guessing correctly 10 times in a row?

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The guesses...

- Let's say we do this, and Joy guesses 10 correctly in a row.
  - Suppose a colleague says, "She just got lucky!"
  - **Can we prove they are wrong?** Any ideas?
- One way of doing it:
  - **Assume for a second** that she really was guessing at random.
  - If that's the case, then **there's a** $< 0.001\%$ **chance** of guessing 10 in a row correctly!
  - Therefore, it seems more likely she *isn't* guessing randomly. The data we observed seems incompatible with our hypothesis.
- We just (informally) did a **Hypothesis Test**
  - Framework for deciding whether or not "due to chance" is a plausible explanation
  - If the event (data, sample) that we observe is extremely unlikely, "due to chance" is not a plausible explanation.

Hypothesis Tests and "P-Values", Informally

- Say we want to **disprove** a hypothesis:
  - e.g., "Joy was guessing randomly, she can't smell Parkinson's disease"
- We are going to calculate something called a **P-value**:
  - **Assuming the hypothesis was true…**

P-value = Probability of what happened in reality (or something more extreme)

- e.g., Assuming Joy was guessing randomly…

P-value = Probability guessing correctly 10 times in a row

- Then we'll **reject the hypothesis if the p-value is small**

Why reject with "small" p-values?

Discuss with Neighbor(s): If the "p-value",

P-value = Probability of what happened in reality (or something more extreme), assuming the hypothesis is true

is small, **why does it make sense to reject the hypothesis?**

- In life, we tend to believe things that happen are pretty "typical" or "likely"… or else they wouldn't happen!
- With a **small p-value**, the hypothesis holding means **"You're special! You observed something really rare!"**
  - We usually don't get that lucky :(So we may conclude that our hypothesis was wrong.
- With a **large p-value**, the hypothesis holding means **"Your observation meets expectations."**
  - In this case, we don't have much reason to doubt our hypothesis.

02:00

Hypothesis Testing Framework

Framework for Hypothesis Testing

Hypothesis Testing is a scientific experiment, and follows the general scientific method.

1. Present research question and identify **hypotheses**
   - **Null** Hypothesis (e.g., Joy is guessing diagnoses randomly)
   - **Alternative** Hypothesis (e.g., Joy is not guessing randomly)
   - We'll express these hypotheses mathematically, *in terms of parameters*
2. Describe **Null distribution**
   - What should we expect to happen if the null hypothesis is true?
3. Obtain data, calculate relevant **Test Statistic**
   - **Test Statistic** = sample statistic based on our data
4. Calculate the **P-value**
   - **P-value** = probability of observing the Test Statistic assuming the Null Hypothesis
5. Use the P-value to **make a conclusion** on the research question

1) Identifying Hypotheses: Informal vs. Formal Hypotheses

- Before the tee shirt experiment, we may have several (informal) hypotheses:
  1. Joy is guessing at random
  2. Joy can smell Parkinson's disease
  3. Joy is psychic
  4. A researcher is somehow cheating to help Joy identify the shirts correctly
- But in order to compare these, it would be helpful to consider a set of hypotheses that:
  - Are mutually exclusive
  - Make specific statements about a **parameter**
  - Do not discuss the specific outcome of the experiment
- **Formal Hypotheses:** Let p denote the true probability that a guess is correct.

$$\textbf{Null Hypothesis: } p = 1/2 \qquad \textbf{Alternative Hypothesis: } p > 1/2$$

- The first informal hypothesis is represented by the Null Hypothesis. The other three are captured in the Alternative Hypothesis.

1) Identifying Hypotheses

- The **null hypothesis** (H_0) is the claim we are testing. It often represents a skeptical perspective or that there is no relationship among several variables.
  - H_0: The chance of guessing correctly is 1/2, or $p = 1/2$.
  - The **null value** = value of the population parameter under the Null Hypothesis (e.g., 1/2)
- The **alternative hypothesis** (H_a) is contrary to the null hypothesis. It is often the theory we would like to prove.
  - H_a: Joy can do better than random guessing, or $p > 1/2$.

1) Identifying Hypotheses

- The Null and Alternative hypotheses are most often statements about the particular value of a population parameter
  - H_0 and H_a are **never** statements about particular values of **sample statistics**. They are **hypotheses** and should be able to be expressed before any observation of data.
    - **Incorrect** H_0: The proportion of correct guesses in 10 guesses is $\hat{p} = 1/2$.
    - **Incorrect** H_a: The proportion of correct guesses in 10 guesses is $\hat{p} > 1/2$.

1) Identifying Hypotheses: Determining the Null Hypothesis

- The Null and Alternative Hypothesis statements are *not* interchangeable:
  - Null hypothesis is (usually) a statement of **equality** for a parameter (e.g., $p = 1/2$)
  - Alternative hypothesis is a statement of **inequality** for a parameter (e.g., $p > 1/2$)
- The null hypothesis should represent the **status quo belief about the parameter.**
  - Would be assumed if no study were conducted, and would be maintained if the study is inconclusive.
- The alternative hypothesis often represents the **claim for which we seek evidence.**
  - It is the only statement we will be able to provide evidence for after our test.
  - e.g., in a criminal trial, we assume innocence (H_0) first. If evidence is compelling ("beyond a reasonable doubt"), then we reject in favor of a guilty verdict (H_a).
- **Note:** Rejecting the Null doesn't mean it's untrue, that's just our conclusion based on the data we have

Framework for Hypothesis Testing

1. Present research question and identify hypotheses
2. **Describe "Null" distribution - What should we expect to happen due to randomness if the null hypothesis is true?**
3. Obtain data, calculate relevant "Test Statistic"
4. Calculate the "P-value"
   - P-value = likelihood of observing the Test Statistic or something more extreme assuming the Null Hypothesis
5. Use the P-value to make a conclusion on the research question

2) Describe the Null Distribution: Likelihood of Observing Sample Statistic

- To compare Null and Alternate Hypotheses, we need to quantify how likely it is to observe a particular sample statistic, *if the null hypothesis were true*.
  - If Joy guesses at random, how many correct guesses do you expect?
  - What is the greatest number of correct guesses you would plausibly expect to see?
  - How likely is is that she would get 7 or more correct guesses, if she's guessing at random?
- To answer questions like these, we need to know the **distribution of the statistic of interest, if the null hypothesis were true**.
- This distribution is called the **Null Distribution** and is the theoretical *sampling distribution* for the statistic if the *null hypothesis* were true.
- We can approximate the Null Distribution using simulation or theory.

2) Describe the Null Distribution: A Model of Tee Shirt Guessing

We can use R to simulate **one experiment** of 10 random guesses by…

- Creating a data frame consisting of correct and incorrect guesses
- Sampling from this data frame *with replacement* 10 times

```
guesses <- data.frame(
  correct = c(0, 0, 0, 0, 0, 1, 1, 1, 1, 1))
guesses
```

```
   correct
1        0
2        0
3        0
4        0
5        0
6        1
7        1
8        1
9        1
10       1
```

```
guesses %>%
  rep_sample_n(size = 10, replace = T)
```

```
   replicate correct
1          1       0
2          1       0
3          1       0
4          1       1
5          1       0
6          1       1
7          1       1
8          1       1
9          1       1
10         1       1
```

This one experiment gives us a $\hat{p} = 6/10 = 0.6$

2) Describe the Null Distribution: A Model of Tee Shirt Guessing

We can use R to simulate **2000 experiments** of 10 guesses by putting in `reps = 2000` (`reps = 1` is the default).

```
guesses %>%
  rep_sample_n(size = 10,
               replace = T,
               reps = 2000) %>%
  group_by(replicate) %>%
  summarize(n_correct = sum(correct)) %>%
  mutate(p_hat = n_correct/10)
```

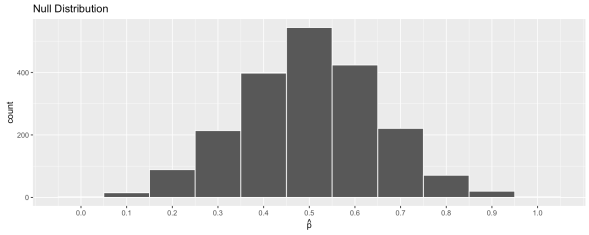
```
# A tibble: 2,000 × 3
   replicate n_correct p_hat
       <int>     <dbl> <dbl>
 1         1         4   0.4
 2         2         5   0.5
 3         3         4   0.4
 4         4         4   0.4
 5         5         8   0.8
 6         6         4   0.4
 7         7         5   0.5
 8         8         4   0.4
 9         9         4   0.4
10        10         4   0.4
# ℹ 1,990 more rows
```

> **Q**: Does this code look familiar?

2) Describe the Null Distribution: Visualizing

- We can use a histogram to visualize the Null Distribution of the sample proportion $\hat{p}$

```
null_stats %>% ggplot(aes(x = p_hat)) + geom_histogram(bins = 11, color = "white")
```

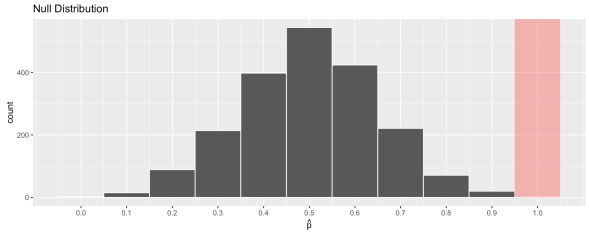


- **Q**: How often would we have observed $\hat{p} = 1.0$?

2) Describe the Null Distribution: Visualizing

- We can use a histogram to visualize the Null Distribution of the sample proportion $\hat{p}$

```
null_stats %>% ggplot(aes(x = p_hat)) + geom_histogram(bins = 11, color = "white")
```



- **Q**: How often would we have observed $\hat{p} = 1.0$?

Framework for Hypothesis Testing

1. Present research question and identify hypotheses
2. Describe "Null" distribution
3. **Obtain data, calculate relevant "Test Statistic"**
   - **Test Statistic = sample statistic based on our data**
4. Calculate the "P-value"
   - P-value = likelihood of observing the Test Statistic or something more extreme assuming the Null Hypothesis
5. Use the P-value to make a conclusion on the research question

3) Calculate the Test Statistic

- We've already done this with Joy's guesses:
  - We found that $\hat{p} = 1$ (she guessed right for 10 out of 10 tee shirts)
- In the context of hypothesis tests, we typically call the sample statistic (e.g., $\hat{p}$) the **Test Statistic**
  - This is the statistic that we actually observed in our data set
  - In comparison, we call the hypothetical statistics in our null distribution **null statistics**

Framework for Hypothesis Testing

1. Present research question and identify hypotheses
2. Describe "Null" distribution
3. Obtain data, calculate relevant "Test Statistic"
4. **Calculate the "P-value"**
   - **P-value = likelihood of observing the Test Statistic or something more extreme assuming the Null Hypothesis**
5. Use the P-value to make a conclusion on the research question

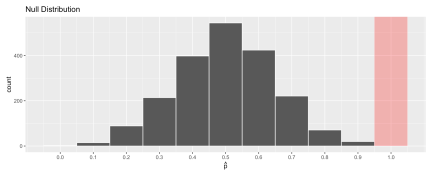
4) Calculate the P-Value

- **Reminder:** Informally,
  - P-value = Probability of what happened in reality (or something more extreme) assuming the null hypothesis is true
- More formally,
  - The **p-value** is the probability of observing a sample/test statistic (e.g., $\hat{p}$) at least as favorable to the alternative hypothesis as the current statistic, if the null hypothesis (H_0) were true.
    - e.g., Joy had $\hat{p} = 1$, so we want the probability of $\hat{p} = 1$ **given** H_0
    - e.g., If Joy had $\hat{p} = 0.5$, we'd want the probability of $\hat{p} \geq 0.5$ **given** H_0
- The p-value quantifies the strength of evidence against the Null Hypothesis.
  - **Smaller p-values represent stronger evidence against H_0.**
  - P-value ≈ 0 means test statistic was unlikely to arise by chance, if the null hypothesis were true.
  - If the p-value is "sufficiently small", we'll reject H_0.

4) Calculate the P-Value

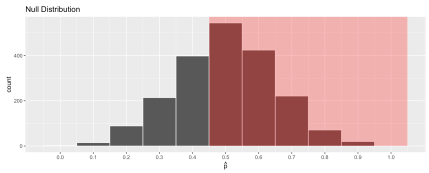
- **Method 1:** Approximate the null distribution using simulation.
  - Then, calculate the proportion of simulated statistics at least as extreme as the test statistic.

For $\hat{p} = 1$, we get a p-value of:




```
1 mean(null_stats$p_hat >= 1)
[1] 0.001
```

For $\hat{p} = 0.5$, we get a p-value of:




```
1 mean(null_stats$p_hat >= 0.5)
[1] 0.641
```


4) Calculate the P-Value

- **Method 2:** We use theory-based tools to create the theoretical null distribution.
  - Then use the model to calculate the theoretical probability of observing a sample statistic as extreme as the test statistic.
  - In the case where Joy gets all 10 guesses right, we have already calculated the p-value as:

$$\text{P-value} = (1/2)^{10} \approx 0.00098$$

Framework for Hypothesis Testing

1. Present research question and identify hypotheses
2. Describe "Null" distribution
3. Obtain data, calculate relevant "Test Statistic"
4. Calculate the "P-value"

- P-value = likelihood of observing the Test Statistic or something more extreme assuming the Null Hypothesis

5. **Use the P-value to make a conclusion on the research question**

5) Making a conclusion using the P-value

Null Hypothesis: $p = 1/2$ **Alternative Hypothesis:** $p > 1/2$

Discuss with Neighbor(s):

1. When $\hat{p} = 1$ (Joy guesses all 10 tee shirts correctly), we found **P-value** < 0.001. Do we reject the Null Hypothesis under this framework? Why?
2. Hypothetically, a $\hat{p} = 0.8$ gives **P-value** ≈ 0.055. Do we reject the Null now? Why or why not?

Answers:

1. Yes, we're rejecting the Null
2. Maybe? But we wouldn't feel as good about it!

- **How small a p-value is small enough to reject the Null?**
  - It really depends on your setting… (and we'll discuss more later)
  - … but people like to reject under 0.05 (this should *NOT* always be the rule!)

02:00

Motivating Example: Can some diseases be detected by smell?

This true story is amazing! I've simplified things here, but I don't want to leave you hanging. The answer is: **yes!** Joy really can smell Parkinson's disease.

When she realized her ability, she and her husband immediately understood the medical significance (she could have diagnosed her husband **10 years earlier** had she known). They reached out to a Parkinson's researcher at the University of Edinburgh.

At first, researchers doubted her, but they decided to test her ability using a similar experiment with tee shirts.

Out of all 12 samples they gave her, she only made one mistake by identifying someone in the control group as having the disease.

However, several months later, that same control individual was diagnosed with Parkinson's.

Researchers and Joy have now found that she can identify more illnesses, including Alzheimer's disease, tuberculosis, cancer, and diabetes. They are working to develop diagnostic tools that use skin-based biomarkers.

Recap: Framework for Hypothesis Testing

Hypothesis Testing is a scientific experiment, and follows the **general** scientific method.

1. Present research question and identify **hypotheses**
   - **Null** Hypothesis (e.g., Joy was guessing randomly)
   - **Alternative** Hypothesis (e.g., Joy can smell Parkinson's disease)
   - We'll express these hypotheses mathematically, *in terms of parameters*
2. Describe **Null distribution**
   - What should we expect to happen if the null hypothesis is true?
   - We'll mainly use simulation to do this, for now
3. Obtain data, calculate relevant **Test Statistic**
   - **Test Statistic** = sample statistic based on our data
4. Calculate the **P-value**
   - **P-value** = probability of observing the Test Statistic assuming the Null Hypothesis
5. Use the P-value to **make a conclusion** on the research question

2) - 4) Null Distribution to P-Value: Full code example

- (2a) Create a data frame reflecting the null hypothesis (here $H_0 : p = 0.5$)
  - Multiple equivalent ways to do this
  - The important thing is that it encodes the assumed **null parameter**

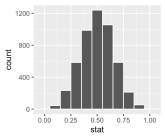
```
guesses <- data.frame(correct = c(0, 0, 0, 0, 0,
                                  1, 1, 1, 1, 1))
```

- (2b) Sample from this data frame (here, *with replacement*) many times, and calculate the **null statistic** (here, the sample proportion) in each sample
  - This simulates what the **sampling distribution** would look like under the **null hypothesis**

```
set.seed(111)
null_stats <- guesses %>%
  rep_sample_n(size = 10, replace = T,  # Creating 2000 samples
               reps = 5000) %>%
  group_by(replicate) %>%  # For each sample...
  summarize(n_correct = sum(correct)) %>%  # Calculate null statistics
  mutate(stat = n_correct/10)
```

- (2c) Visualize the null distribution, check that the center makes sense.

```
null_stats %>%
  ggplot(aes(x = stat)) +
  geom_histogram(bins = 11,
                 color = "white")
```



- (3) Here, we were given the value of the test statistic: $\hat{p} = 10/10 = 1$.
- (4) Calculate your p-value.
  - In other examples, you may need to calculate `test_stat` from a data frame with the sample.

```
test_stat <- 1
mean(null_stats$stat >= test_stat)  ## p-value
```

```
[1] 6e-04
```


Next time

- More hypothesis testing!
  - Practice framing research questions in the hypothesis testing framework
    - Particularly, generating null distributions
  - 1-sided vs 2-sided tests